Project Proposal

SignBridge-TR: A Real-Time, Bidirectional AI-Assisted Translation System between Turkish Sign Language, Turkish Text and Speech for Hearing-, Speech- and Visually-Impaired People

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Introduction

SignBridge-TR in brief. SignBridge-TR is a browser application that helps Deaf, speech-impaired and blind people talk with people who do not know Turkish Sign Language (TİD), in two directions: (A) a TİD user's signs are recognised from the camera and turned into Turkish text and speech; (B) Turkish speech or text is turned into TİD signs performed by a 3D avatar. The first version, the minimum viable product (MVP), recognises isolated TİD signs separated by user pauses and supports constrained conversion between gloss sequences and Turkish sentences within the selected vocabulary and scenarios. Turkish-to-sign output uses a predefined avatar animation library. Recognition and translation of unrestricted continuous TİD signing are outside its scope. The recognition model is trained on existing public datasets (the Dataset; see Key terms); supplementary recordings are only a conditional option (Methods 1).

Key terms used in this proposal.

Dataset	AUTSL, or BosphorusSign22k if AUTSL cannot be downloaded or used under its terms.	**WER**	Word error rate of speech recognition.
Gloss	Written label of a sign, e.g. [OKUL] for "school".	**BLEU / chrF**	Automatic sentence-similarity scores.
Landmarks	MediaPipe landmarks (keypoints): hand, body and face points estimated per frame.	**SUS**	System Usability Scale (0–100).
MVP	Minimum viable product: the first, limited version.	**WCAG 2.2 AA**	W3C accessibility guidelines, level AA.
Signer-independent	Tested on people whose signs were not used in training.	**KVKK**	Turkish Personal Data Protection Law No. 6698.
Top-1 / top-5	Correct class is the first / among the five best guesses.	**ONNX / TF.js; glTF**	Browser model formats; 3D animation format.

Subject area and problem. Our project belongs to assistive technology and draws on computer vision, natural language processing and human–computer interaction. Its topic is automatic translation between a sign language and a spoken language in both directions: recognising signs in video and turning them into text and speech, and turning speech or text into signs performed by a virtual character (sign language production).

According to the World Health Organization, about 430 million people need rehabilitation for disabling hearing loss, a number expected to pass 700 million by 2050 [1], and at least 2.2 billion people have a near or distance vision impairment [2]. In Türkiye, many Deaf people grow up with TİD as their first language; TİD has its own vocabulary and grammar and is not Turkish done with the hands. Article 15 of Law No. 5378 provides for the creation of a Turkish Sign Language system to support the education and communication of people with hearing impairments [3], and since 2017 the Ministry of Family and Social Services has published the Current TİD Dictionary online, covering the 2,000 most frequent words [4].

Even so, everyday communication between Deaf and hearing people still relies on interpreters, who are few and must be booked days ahead, or on written notes, which are slow and often fail because many Deaf adults read Turkish as a second language. At a hospital desk, a pharmacy, a bank or a classroom, the hearing person rarely knows TİD. Speech-impaired people who sign can hear but cannot answer aloud, and blind people can neither see signs nor read notes, so a blind person and a Deaf person have almost no direct way to talk.

The problem we want to solve is that, among the products and studies we reviewed (Table 4), we found no tool that TİD users and non-signers, including blind and low-vision people, can pick up at any moment to talk in both directions. This matters because communication barriers turn directly into unequal access to healthcare, education, work and public services, and finding an interpreter for every short conversation is not realistic.

Current research and development. *Sign language recognition* relies on deep learning on RGB video or body and hand keypoints. The two large public TİD datasets are AUTSL (226 signs, 43 signers, 38,336 isolated samples with RGB, depth and Kinect skeleton data) [5] and BosphorusSign22k (744 signs, 6 signers, 22,542 samples) [6]. In the 2021 ChaLearn challenge, multi-modal models such as SAM-SLR exceeded 98% top-1 accuracy on AUTSL [7]; these are single-sign benchmark results, not live-conversation performance. *Body tracking:* Google's MediaPipe [8] estimates hand, body and face landmarks and is designed to run in real time in the browser or on a phone, so video need not be sent to a server; its speed on our devices is measured (Methods 4). *Translation and production:* neural sign language translation was first shown on German Sign Language data [9], and Progressive Transformers generated skeleton motion from text [10]. Google announced SignGemma in May 2025 as a model for translating sign languages into spoken-language text, best on American Sign Language (ASL) to English [11]; Hand Talk translates text and speech into Brazilian Sign Language and ASL with avatar translators [12]; Engelsiz Çeviri translates web pages, subtitled videos, PDFs and printed material into TİD with a 3D avatar or a human model [13]. *Speech and language:* multilingual models such as Whisper transcribe Turkish speech [14], and large language models (LLMs) handle Turkish morphology and rephrasing, so we use them to convert between Turkish sentences and gloss sequences.

The gap. Based on our literature review and on the websites of the products in Table 4, checked on 21 October 2026 (SignGemma through Google's 2025 announcement), we identified four gaps:

(1) To our knowledge, products in Türkiye translate content into TİD in one direction only; Engelsiz Çeviri lists a two-way tool, from speech and text to sign language and back, as under development [13].

(2) Academic work on TİD mostly stops at recognition accuracy; very few models trained on AUTSL or BosphorusSign22k have been turned into end-to-end conversation systems and tried with Deaf users in a real setting.

(3) Support for blind users was not found in the tools we reviewed: the sign language tools in Table 4 assume the user can see the avatar or read captions, and camera-based tools assume the user can aim the camera. We found no reviewed system that lets a blind person and a Deaf person talk to each other.

(4) Models from large companies focus on ASL, whereas TİD has its own vocabulary and grammar and therefore needs its own data and rules.

Product concept: how SignBridge-TR fills the gap. SignBridge-TR is a web-based progressive web app (PWA) that runs on phones, tablets and computers and translates in real time in both directions (Figure 1):

1. **Direction A – from signs to speech:** camera frames are turned into hand, body and face landmarks on the device; a landmark-based recognition model recognises the TİD signs (the model type is chosen by comparing candidates; see Methods); an LLM builds a proper Turkish sentence from the gloss sequence, which is shown in large letters and read aloud.
2. **Direction B – from speech or text to the avatar:** Turkish speech is transcribed; the LLM, together with our TİD rule layer, converts the sentence into a gloss sequence; each gloss matches an animation clip in our sign library, and the 3D avatar plays the clips as a smooth sequence with a caption and adjustable speed. Words not in the library, such as names, are fingerspelled.
3. **Accessibility layer for blind and low-vision users (proposed):** every result is read aloud and passed to screen readers (VoiceOver, TalkBack, NVDA); a small set of in-app voice commands, with keyboard and touch alternatives, controls the main functions; a sound tells the user when the other person's hands are fully in the frame, and a confirmation signals that a message has reached the other side. These blind-specific features are proposed, pending a team/advisor scope decision (it may affect RQ4, O6 and O7), and are not an agreed MVP commitment; general accessibility (keyboard, readable text, captions) stays in scope.

A key design decision is that the AI does not generate the avatar's arm and hand movements from scratch: an intermediate layer links glosses to a prepared animation library, so the avatar's signs can be verified and the project stays feasible in two semesters (fully generative motion is future work). **What we design and implement versus reuse:** we reuse MediaPipe for landmark extraction, existing speech-to-text, text-to-speech and LLM models, and the Dataset for training the recognition model; supplementary sign recordings are only a conditional option (Methods 1). We design and implement the TİD recognition model and its training pipeline, the landmark extraction and preprocessing pipeline, the TİD lexicon, the Turkish ↔ TİD rule layer, the animation library and clip-blending engine, the real-time conversation engine (FastAPI + WebSocket) and the accessibility layer.

Major risks and contingency plans. Table 1 lists the risks that could affect the project and our response to each.

Table 1 – Risk register (L / I = likelihood / impact: H high, M medium, L low)

Risk	L / I	Mitigation and contingency
Not enough TİD training data for the target signs	H / H	Train on the Dataset first, using MediaPipe landmarks extracted from its videos; augment (mirroring, time shifts, landmark noise, variable pauses); fit the MVP vocabulary and scenarios to the coverage audit; supplementary recordings only under the conditions in Methods 1.
The Dataset cannot be downloaded or used as planned	M / H	Check completeness and integrity of the full download, the class mapping and the terms in the feasibility check; keep preprocessing dataset-independent; a parallel BosphorusSign22k licence request awaits team approval.

Risk	L / I	Mitigation and contingency
The model fails for new people, lighting or backgrounds	M / H	Test on unseen signers; examine lighting, background, occlusion and fast movement in the landmark-quality check; report live-webcam results separately; give framing guidance; confirm sentences before speaking.
Wrong grammar conversion between Turkish and TİD	M / H	Check the rule layer and example prompts against the Current TİD Dictionary (and a TİD interpreter if one can be reached); limit output to library glosses; fall back to fingerspelling; rate held-out test sentences (Methods 5).
Avatar signs are hard to understand (finger shapes, facial expressions)	M / H	Base clips on the Current TİD Dictionary and the Dataset's reference videos, reviewed by a native signer if available; use finger bones and facial expressions; test comprehension with Deaf TİD users where reachable; a reference-video comparison is reported separately and does not replace it.
Producing the avatar library takes longer than planned	M / H	Review clip source and usage terms, candidate methods and effort per clip in a pilot (avatar approach research) before the Statement of Work; the ≥300 target is kept, and any change is decided with the advisor.
Total delay above the target	M / M	On-device model (ONNX Runtime Web / TF.js, compressed), cached phrases, smaller model if needed; devices benchmarked before the Statement of Work; server-assisted classification helps only if classification is the bottleneck.
No TİD interpreter or too few Deaf and blind participants	M / M	Initial contact has been made with the university's disability support unit and a response is pending; list at least three institutions (e.g. Deaf and blind associations), contact at least two and follow up; otherwise run only the simulated pre-tests, report the target-user measures as not measured and agree participant numbers with the advisor.
Ethics committee approval takes longer than planned	M / M	Decide the recording question with the advisor, then submit the application, protocol and consent forms early in the first semester (review may take 2–3 months); if approval is late, shorten the April–May evaluation and report a limitation.
Privacy of face and hand video (personal data under KVKK)	L / H	Raw camera video never leaves the device; audio, landmarks and text only as in the data-flow table (Methods 6); a test protocol with fictional scenarios and no real patient data is proposed, not yet agreed; written, explicit consent; no participant test before ethics approval.
AI services become paid or unavailable	M / L	Keep open-source alternatives (local Whisper, an open-source LLM, Piper TTS) behind the same service interface; cache responses.

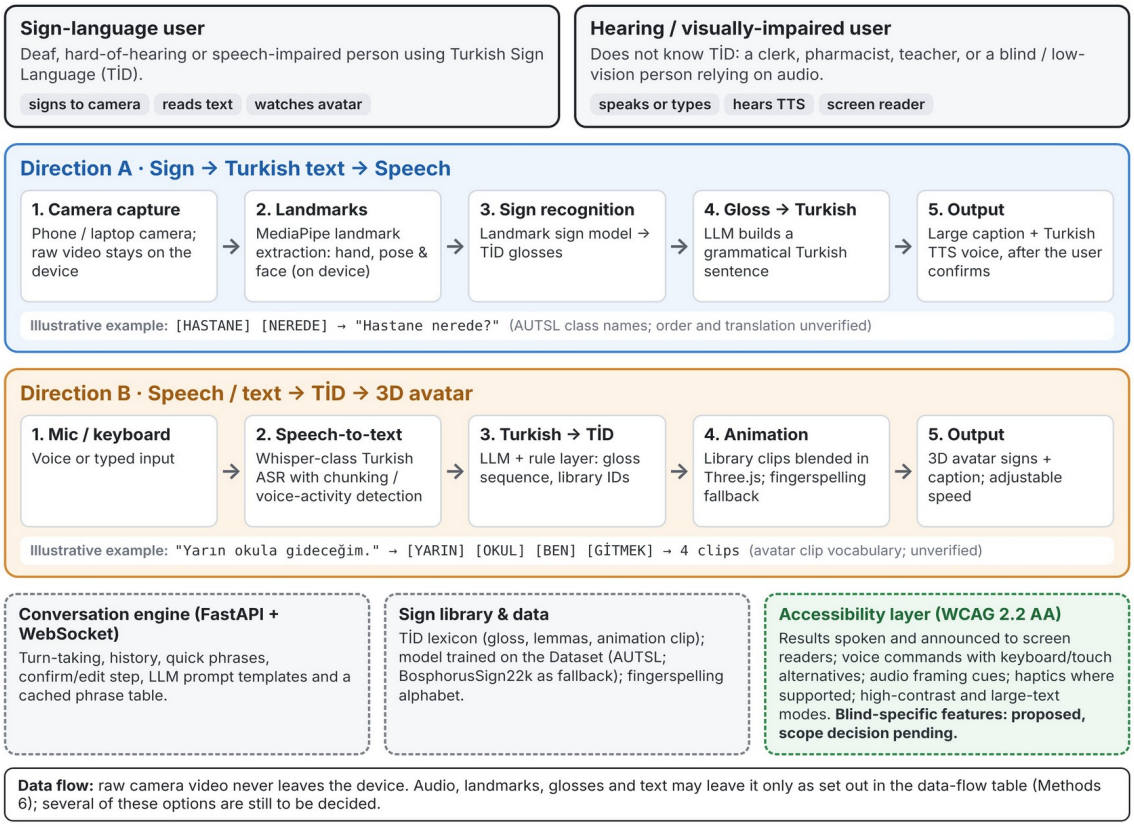


Figure 1 – SignBridge-TR system architecture: two real-time directions share one conversation engine and one accessibility layer; raw camera video never leaves the device, and the data that may leave it are listed in Methods 6. Gloss examples are illustrative and linguistically unverified; Direction A uses AUTSL class names, and Direction B uses the separate avatar clip vocabulary.

Why we are suitable for this project. Our six-person team covers machine learning (PyTorch), web development (React/Next.js), back-end development (Python, FastAPI), 3D graphics (Three.js, Blender) and user-experience design. Support from a TİD interpreter and early involvement of Deaf and blind users are desirable but not yet confirmed. Technical development can proceed without them, but the target-user evaluation (O4 user measures and the second O7 stage) depends on reaching suitable participants (Table 1). We chose this topic to apply the AI methods we have learned to a problem that directly affects people's daily lives.

Contribution to the UN Sustainable Development Goals. The project contributes most directly to **SDG 10, Reduced Inequalities** [15] (target 10.2: social and economic inclusion of all irrespective of disability) by removing an everyday communication barrier, and indirectly to **SDG 4, Quality Education** (target 4.5: students can talk to classmates and teachers without waiting for an interpreter), **SDG 8, Decent Work and Economic Growth** (target 8.5: easier communication at work) and **SDG 3, Good Health and Well-being** (target 3.8: easier access to healthcare, where misunderstandings are most dangerous).



UN Sustainable Development Goals [15]; the project contributes directly to SDG 10 and indirectly to SDGs 4, 8 and 3.

Aim and Objectives

Aim. To design, develop and test SignBridge-TR, a real-time, two-way application that translates Turkish Sign Language into Turkish text and speech, and Turkish speech or text into TİD signs performed by a 3D avatar, designed so that Deaf, speech-impaired and blind people (blind-specific features: scope pending) can hold short conversations without an interpreter; this is evaluated as task completion without researcher assistance, in simulated pre-tests and, where volunteers can be reached, with target users (O7).

Objectives. Each objective has a measurable success criterion that will be checked at the end of the project (Table 2).

Table 2 – Objectives and success criteria

#	Objective	Success criterion (how it will be measured)
O1	Recognise single TİD signs from a phone or computer camera in real time.	MVP vocabulary selected from the Dataset's sign classes (AUTSL: 226; BosphorusSign22k: 744), with priority for health, pharmacy, public services and daily conversation, fixed after the coverage audit using training and validation data only. ≥85% top-1 and ≥95% top-5 accuracy on the Dataset's signer-independent test split; ≥15 frames per second (fps) of landmark processing in the browser on a mid-range laptop (a laptop target, not evidence for phones). These are benchmark results; live-webcam accuracy is reported separately.
O2	Build grammatical Turkish sentences from recognised gloss sequences.	≥80% (point estimate, with its 95% interval) of a held-out MVP test set of 100 gloss sequences, separate from the development examples, judged "correct meaning and correct grammar" by two native Turkish speakers with a written rubric; BLEU/chrF reported against reference sentences. The result covers this test set only.
O3	Transcribe Turkish speech in streaming mode.	Planned controlled test: word error rate ≤15% on 200 sentences recorded in a quiet room (audio source not yet chosen; Methods 5); WER for other speakers, devices, distances and background noise reported separately, without a target.
O4	Translate Turkish into TİD and show it with a 3D avatar.	Animation library of ≥300 signs plus the full Turkish fingerspelling alphabet; ≥75% of signs correctly recognised and ≥70% of 50 test sentences understood by Deaf TİD users. A reference-conformance check against the Current TİD Dictionary video (≥75% of signs passing) is reported separately and does not replace these user measures; without suitable TİD users they are reported as not measured and those O4 criteria are not verified (Methods 5).
O5	Work fast enough not to break the flow of a conversation.	Median system time from the end of the last sign (including pause detection) to the start of spoken output ≤2.0 s; median time from the end of speech to the avatar starting to sign ≤2.5 s. Time a user spends confirming a sentence, and manual-mode interaction time, are excluded and reported separately (event definitions in Methods 4).
O6	Make every flow in the MVP accessible.	Compliance with WCAG 2.2 level AA [16] (automated axe-core scan plus manual checks); all four persona flows completed from start to finish with VoiceOver/TalkBack and with the keyboard only.
O7	Evaluate usability in two stages.	(1) Pre-tests under simulated conditions in the defined service scenarios (system behaviour, interface problems); (2) if suitable volunteers can be reached, an additional evaluation with target users: Deaf/hard-of-hearing TİD users, blind users and hearing non-signers. Participant numbers, distribution and sampling are agreed with the advisor before testing. ≥80% task completion and a mean SUS score ≥70 [17], reported per stage and group with actual numbers and limitations.
O8	Protect users' privacy.	Raw camera video never leaves the device (checked by code review and network-traffic inspection); audio, landmarks and text leave it only as set out in the data-flow table (Methods 6); written, explicit consent from every evaluation participant, collected only after ethics committee approval.

User Personas of the Product

Target audience. We expect three main user groups: Deaf and hard-of-hearing people who use TİD; speech-impaired people who hear but communicate in TİD; and blind or low-vision people who need to talk to a TİD user. The hearing people on the other side who do not know TİD are also users: hospital and municipal staff, pharmacists, teachers, classmates and family members. On the institutional side, hospitals, pharmacies, municipalities, schools and universities are possible users. Users differ widely in age and comfort with technology, so the interface has to work without any training. From these groups we created four personas (Table 3).

Table 3 – User personas

Persona	Occupation and demographics	Background story	Challenges	What they need to be able to do
Ayşe Yılmaz (Deaf, uses TİD)	Graphic designer at a printing company; 34, female, İzmir. Deaf from birth, TİD is her first language; reads Turkish at a basic level; uses her phone a lot, especially for video calls in TİD.	Went to a school for the Deaf. For hospital appointments she books an interpreter days in advance or takes her sister. When she gets sick suddenly she writes notes, and staff reply by speaking or with long, complicated sentences.	Cannot find an interpreter in emergencies; long written Turkish is hard to follow; staff lose patience; relying on family for health matters takes away her privacy.	Sign comfortably in front of her phone and have it read aloud; see replies in TİD, not only as text, and replay them; start within a few seconds without setup.
Mehmet Kaya (hearing, does not know TİD)	Pharmacist with his own neighbourhood pharmacy; 52, male, Konya. Hearing, does not know TİD; moderate technology skills (uses a tablet for prescriptions).	A few Deaf customers come every month. He explains how to take medicine by writing and gesturing, and worries that a misunderstood "twice a day, after meals" could hurt someone.	Does not know sign language; written instructions are sometimes misread; has to hurry when there is a queue.	Speak normally and have it signed correctly, including numbers and times; understand the customer's question; use the app without learning anything first.
Zeynep Arslan (blind student; proposed, scope pending)	Third-year psychology student; 21, female, Ankara. Totally blind since childhood; very good with VoiceOver; hearing.	She is in the same project group as her Deaf classmate Emre. They can text to make plans, but face to face they can only talk through a friend who knows a little TİD.	Camera apps assume the camera is aimed correctly; many apps work poorly with screen readers; she cannot tell whether Emre is in frame or whether her message got through.	Use the whole app by voice or with a screen reader; get spoken framing guidance; hear what Emre signs; know that her spoken reply was shown to Emre.
Can Demir (speech-impaired, uses TİD)	Warehouse technician; 28, male, Bursa. Hears well but cannot speak because of a congenital speech disorder; learned TİD at a special-education school.	At a café, shop or bus terminal he types what he wants on his phone and shows it; in a busy queue this is far too slow and people answer before he finishes.	Typing takes time; people do not bother to read notes; text-to-speech apps want typed text instead of signs.	Have what he signs read out at once in a voice he chose; check the sentence before it is spoken; save frequent sentences and reuse them with one tap.

Happy Paths of the User Personas

Each persona has one happy path, from the persona's goal to the completed task: (a) Ayşe uses the two-way mode at a hospital desk; (b) the flow starts from the hearing side with Mehmet, and numbers and times are signed explicitly; (c) Zeynep (proposed path, scope decision pending) uses a small set of in-app voice commands, VoiceOver and keyboard controls, helped by spoken framing guidance; (d) Can uses the sign-to-speech mode, confirms the sentence before it is spoken and saves a frequent phrase. Haptic feedback is used only where the platform supports it, and the app is launched from the installed PWA's home-screen icon. The wireframes of the four paths (Figure 2 of v1.0, corrected) are in the Supporting Technical Notes, at a readable size, and are developed further in the Statement of Work, where the Guidelines grade Happy Paths.

Expected Outcomes/Deliverables

At the end of the project we will deliver:

1. **SignBridge-TR application:** a PWA with four modes (Sign → Voice, Voice → Sign, Two-way conversation, Quick phrases) that runs in desktop and mobile browsers; source code on GitHub with installation and user documentation.
2. **TİD sign recognition model:** a landmark-based model exported to run in the browser (ONNX / TF.js), with training code and an evaluation report (accuracy, confusion matrix, results on unseen signers, baseline comparison).
3. **TİD sign animation library:** at least 300 signs and the fingerspelling alphabet in glTF format; each sign's gloss, Turkish equivalents, domain and clip file kept in a JSON lexicon.
4. **Turkish ↔ TİD translation module:** prompt templates, the rule layer and held-out Turkish–gloss test sets for both directions (100 sequences for O2), checked against the Current TİD Dictionary (and a TİD interpreter if one joins the project).
5. **Dataset audit and landmark pipeline:** an audit of the Dataset (classes, scenario coverage, signers, splits) with reproducible landmark extraction and preprocessing scripts and a data card; landmarks are not redistributed unless the licence allows it.
6. **Statement of Work and design documentation:** functional and non-functional requirements, architecture, use-case and UML diagrams, API definitions and interface prototypes.
7. **Test plan and test reports:** unit and integration tests, model comparisons, latency and device benchmark results and a WCAG 2.2 AA audit report.
8. **Usability evaluation report:** protocol; results of the simulated pre-tests and, if held, of the target-user evaluation, reported separately (actual participant numbers and groups, task completion, SUS, limitations); recommendations for future work; the ethics committee approval.
9. **Post-implementation report, demo video, poster and presentation**, together with the sprint records kept in Jira and Confluence.

Related Research and Existing Solutions

Table 4 – Comparison with existing solutions

Solution	Language	Sign → text / speech	Text / speech → sign	Real-time two-way	Blind-user support
Engelsiz Çeviri [13]	TİD	Planned (under development)	Yes (avatar or human model)	Planned (under development)	Not described
Hand Talk [12]	Libras, ASL	Not described	Yes (avatars)	No	Not described
SignGemma [11] (announced 2025)	ASL	Yes (to English)	No	No	Not described
AUTSL / BosphorusSign22k research [5]–[7]	TİD	Single signs, offline tests	No	No	No
SignBridge-TR (proposed)	**TİD**	**Yes (text + speech)**	**Yes (3D avatar)**	**Yes**	**Proposed: voice, screen reader, spoken guidance (scope decision pending)**

Research Questions and Hypotheses

RQ1. How accurately can a landmark-only model trained on the Dataset recognise TİD signs from people it has not seen before, in real time, on ordinary devices?

RQ2. Can an LLM supported by a TİD-specific rule layer convert between a Turkish sentence and a TİD gloss sequence without losing the meaning (judged by TİD users or, if they cannot be reached, by raters using the Current TİD Dictionary)?

RQ3. Can Deaf TİD users understand avatar signs built from the library well enough to complete a short service conversation?

RQ4. Can a blind user talk with a Deaf person using only voice and a screen reader? (Depends on the open scope decision.)

Hypothesis 1: the landmark-based Transformer reaches at least 85% top-1 accuracy on the MVP vocabulary for people not seen in training; the final model is chosen by comparison with baselines and a graph network, not assumed in advance. **Hypothesis 2:** adding the rule layer to the LLM, with the same model version, prompt template and examples, significantly increases the share of meaning-preserving conversions compared with the LLM without the rule layer, tested on the same sentences (see Methods). **Hypothesis 3:** in each evaluation stage, participants complete at least 80% of the scenario tasks and the mean SUS score is at least 70; stages and groups are reported separately.

Methods

Development process. We follow Scrum as defined in the Senior Project Handbook: two-week Jira sprints with one epic per sub-system, Agile Poker estimation (1-2-3-5), daily stand-ups, a sprint review with our advisor as Product Owner, a retrospective, sprint planning and at least one technical meeting per sprint whose action points become Jira issues; records are kept in Confluence, and the team-leader role rotates. Code is integrated with GitHub Flow: one branch per Jira issue (story/, task/ or bug/ plus the issue ID), pull-request reviews, release tags and continuous integration with GitHub Actions. Technical ownership has six areas: back-end and LLM; computer vision/ML; data and lexicon; avatar and animation; front-end and accessibility; speech, evaluation and testing.

Technical approach. The plan follows the order in which the work depends on it: (1) data and the feasibility check, (2) the recognition model, (3) the other components, (4) device performance and latency, (5) evaluation, (6) data protection and ethics, and (7) review points. Detailed protocols are in the accompanying Supporting Technical Notes; the criteria below do not depend on them.

1. Data and feasibility check

The model is trained on the Dataset under its licence terms, for academic, non-commercial use and with citation [5], [6]. The current training plan prioritises existing datasets, starting with AUTSL and considering BosphorusSign22k where suitable and permitted. If the coverage audit identifies gaps that cannot be addressed through these datasets or reasonable scope adjustments, supplementary sign recordings may be considered. Any new data collection will require advisor agreement on the revised scope, workload and schedule, together with the necessary ethics approval, participant consent and data-use arrangements before collection begins. Supplementary recordings would keep the signer-independent split and the rule against selection on test results. AUTSL access was requested on 20 October 2026, and an AUTSL author replied. Access instructions were provided on 21 October 2026. A team member subsequently reported downloading the data. Decoding and label matching were verified for the supplied sample files; completeness and integrity of the full dataset remain to be checked. The label files match the challenge release of AUTSL [18]: 36,302 samples of 226 classes, split by signer into 28,142 training (31 signers), 4,418 validation (6) and 3,742 test (6) samples with no signer shared between splits, unlike the 38,336 samples reported in [5]. Only the RGB videos are used (RGB and depth files of one sample ID can be paired, but no RGB-D model is planned); their 512×512, 30 fps format says nothing about our landmark-processing rate. Access gives no right to redistribute the data or derived landmarks or animations. Pipeline: RGB videos → MediaPipe landmarks (not the Dataset's Kinect skeleton data) → class labels → glosses and Turkish equivalents in our lexicon (checked by hand) → recognition model. Training and browser use the same MediaPipe version and point schema, not chosen yet (legacy Holistic: 543 landmarks [19]; Holistic Landmarker: 553 [20]).

Feasibility check (late October – early November 2026, before the Statement of Work scope is fixed on 13 November): (a) completeness and integrity of the full download and the terms of use; (b) class list and gloss mapping; (c) a scenario coverage audit listing, per persona scenario, the required glosses, exact class matches, uncertain matches (synonyms or compounds needing validation) and missing glosses; coverage = exact matches ÷ required glosses, fingerspelled words excluded. The class list contains, for example, hospital, doctor, medicine, pharmacy, hour, help and tomorrow but not appointment, stomach, pain or go, and one hour class does not cover numbers and times; scenarios are fixed after the audit. A missing class is a vocabulary gap for our scenarios, not a sign of an incomplete download (checked in (a)). (d) landmark quality and preprocessing time on sample clips; (e) a first MediaPipe web benchmark on the test devices; (f) offline and browser landmarks compared on the same clips. Broader training follows from December; decision thresholds are agreed with the advisor.

2. Recognition model: what is trained and why

Training or adapting a model is needed because the MVP must map browser-extracted landmarks to TİD sign classes, and no ready-to-use model that does this for TİD in the browser has been identified so far. Three options are compared on the same data, landmarks and splits:

- *Baselines* trained from scratch, such as a small recurrent (GRU/LSTM) classifier, to show whether the task is learnable.
- *Models trained from scratch:* a lightweight Transformer encoder and a spatio-temporal graph network on 32–64-frame windows.
- *Transfer learning* from a pose-based pretrained model such as OpenHands [21], only if its pose format matches our landmarks, its licence allows our use and it can be exported to the browser; this has not been checked yet.

Options are compared on accuracy, model size, browser export (ONNX/TF.js) and inference time on the Dataset's signer-independent split [5]. Vocabulary and model are chosen on training and validation data; the test split is used once, and the vocabulary is not changed afterwards; O1 and Hypothesis 1 use the same MVP classes. Live-webcam results from the evaluation are reported separately as exploratory; evaluation data are not used for training. The O2 sentences evaluate the language module only.

Sign boundaries (proposed, not yet implemented or validated): a single short pause is not taken as a sign end, since some signs contain holds; automatic segmentation would combine hand motion, return to a rest pose and pause duration, with thresholds set in pilot tests. If it proves unreliable, a manual mode in which the user starts and ends the capture window for one sign is a possible fallback (temporary on-device processing: no video saved or sent, no training data), subject to the feasibility results and the advisor's scope review. An accessible "done / finish input" control (button, keyboard, screen-reader label; voice only as an alternative) marks the end of the whole sign sequence, not of one sign, and differs from the confirm/edit step before speaking. Training adds variable pauses, as Dataset clips start and end with the hands down.

3. Language, speech, avatar and accessibility components

- **Gloss → Turkish:** an LLM with example prompts answers in a fixed JSON format; frequent sentences come from a cache.
- **Turkish → TİD gloss:** the sentence is analysed morphologically and reduced to word roots (e.g. with Zemberek); the LLM orders the glosses with rules documented in our lexicon design, to be reviewed by a TİD expert where available; each word is matched to a library sign, then to synonyms, and otherwise fingerspelled.
- **Avatar animation:** a human-like avatar with finger bones and facial expressions; the library is separate from the recognition vocabulary and not limited to the Dataset's classes. Clip source and production method (candidates such as manual animation or retargeting landmarks from Dataset videos; none chosen or validated) are selected in the avatar approach research after checking usage terms, quality and effort in a pilot, and agreed with the advisor, without our own motion-capture recordings; clips are saved as glTF and blended in Three.js.
- **Speech:** a Whisper-class Turkish speech-to-text model, used with audio chunking and voice-activity detection because it does not stream natively, and a Turkish text-to-speech engine with several voices.
- **Accessibility:** ARIA live regions for screen readers, the Web Speech API for voice commands with keyboard and touch alternatives, framing alerts based on hand-landmark visibility, and haptic feedback through the Vibration API where supported (Android; on iOS, where web vibration is unavailable, audio cues are used instead).

4. Device performance and latency

The ≥15 fps target in O1 is a laptop target; no device is called supported until tested. On each test device we measure the camera frame rate, the landmark-processing rate, the classification time per window and end-to-end latency, plus missing or occluded hands and dropped frames in fast movement, and record hardware, operating system, browser, model size and conditions. The official MediaPipe guide gives no frame-rate or hardware figures [20], so minimum requirements come from the first benchmark.

Bottlenecks: if landmark processing misses the target, lighter MediaPipe settings and lower resolutions are tried; otherwise the device is not supported. Server-assisted classification reduces only the classification cost, as landmark extraction stays on the device; it would send landmark sequences, never raw video, and is not decided. A bandwidth estimate (assumption only) is in the Supporting Technical Notes.

Latency (O5): in automatic mode the sign-end event is the last frame of the final sign before the rest pose or pause; the time to spoken output includes pause detection, classification, the LLM and TTS. In the manual fallback, processing starts

when the user ends the capture window; its results and the user's interaction time are reported separately and not used alone to claim O5 (how O5 applies to it is an advisor decision). From the end of voice activity, the time to the avatar's first frame includes chunked speech-to-text, the LLM and clip loading. Stage times are recorded.

5. Evaluation plan

Recognition (O1): top-1/top-5 accuracy and a confusion matrix on the signer-independent test split, plus the offline-versus-browser comparison.

Gloss → Turkish (O2): 100 held-out gloss sequences, separate from the prompt and rule development examples and varied in domain, sentence type (statements, questions, negation, numbers, times) and unseen phrasing, are rated by two native Turkish speakers with a written meaning-and-grammar rubric. Cohen's κ [22] and percent agreement use the independent ratings before discussion; the agreed ratings give the success rate, whose point estimate is compared with ≥80% and reported with its 95% Wilson interval [23]. It describes this test set only, not a per-user failure rate; task success is measured in O7.

Turkish → gloss (Direction B, RQ2): a separate held-out set of Turkish sentences is converted to glosses, and each output is rated against references with listed acceptable alternatives for preserved meaning, numbers, times, questions and negation, and for acceptability; exact match or word order alone is not proof of correct TİD. References come from a TİD interpreter or fluent signer if available, otherwise from the team using the Current TİD Dictionary (reported as not expert-validated). Set size and any threshold are agreed with the advisor.

Hypothesis 2: the LLM alone and the LLM with the rule layer convert the same held-out sentences; the rule layer is the only difference (same model version, prompt template, examples and settings). Outputs are rated meaning-preserving or not, and the paired results are compared with McNemar's test [24], reporting the difference and its uncertainty.

Speech (O3): the 200-sentence quiet-room test is a controlled baseline for the ≤15% target; WER for other speakers, devices, distances and background noise is reported per condition. The audio comes from a licensed Turkish speech corpus or from recordings within the approved ethics process; this choice is open.

Avatar (O4): three kinds of evidence are kept apart. (i) *Sign recognition by Deaf TİD users:* clips are shown without captions or other cues, participants give the meaning in their preferred way, and answers are scored against a prepared key of acceptable meanings; rate = correct clip–participant answers ÷ all valid answers, with replay and missing-answer rules set before testing; O4's "≥75% of signs" uses this answer-level rate (a per-sign rule is an advisor decision). (ii) *Reference conformance:* two reviewers independently compare each clip with its dictionary video on handshape, orientation, location, movement and applicable non-manual features; reviews by the clips' makers are not independent evidence, and the check does not measure comprehension. (iii) *Sentence comprehension:* the 50 sentences are shown without Turkish text; an answer is understood if its key content (who, what, when, numbers) matches the meaning key without contradiction; rate = understood answers ÷ all valid answers. Without suitable TİD users, (i) and (iii) are reported as not measured and those criteria are not verified; (ii) does not replace them. κ and percent agreement are computed as in O2.

Overall: the evaluation combines numerical measurements (top-1/top-5 accuracy, BLEU/chrF and human ratings, WER, latency, axe-core and manual WCAG checks) with usability tests in the four persona scenarios, each followed by the SUS [17] and short interviews, in two stages: simulated pre-tests, then target users if suitable volunteers can be reached. Stages are reported separately with actual numbers, groups and limitations; simulated conditions are not equivalent to lived disability experience, so simulated results alone do not verify usability for target users, and small samples are not generalised. In O7 a task counts as completed only without researcher assistance.

6. Data protection and ethics

Data flow of the application (data-flow table)

Data	Processed	May leave the device	Processed by	Retention	Open decision
Camera video	On the device (MediaPipe)	Never	—	Not stored	—
Microphone audio	Speech-to-text; browser voice commands	If an external speech service is used; browser speech recognition is server-based by default unless on-device processing is available [25]	Speech provider or browser vendor	Not stored by the app	Local or external; provider
Landmark sequences	On the device	Only if server-assisted classification is chosen	Own back-end	Not stored	Whether sent
Glosses and Turkish text	Back-end, LLM, TTS	For sentence generation, Turkish → gloss and speech output	Own back-end; local or external LLM/TTS	To be set before the evaluation	Local or external; retention

Speech and text: before any external speech-to-text, LLM or TTS service is chosen, its data-processing terms are assessed against KVKK; no provider, retention period or guarantee is assumed. We propose scripted, fictional service scenarios and no real patient data in tests; the team has yet to confirm this. *Research data:* evaluation data are research data, not training data; any supplementary sign recording (Methods 1) is a separate collection with its own approval and consent. Whether sessions are recorded on video or audio is not decided, while our draft consent forms assume both; this is decided with the advisor, and the application, protocol and consent forms follow it. *Ethics:* the evaluation involves human participants and their disability status (health data under KVKK), so it needs ethics approval; we apply with our advisor in the first semester, as the review may take two to three months, and no test with participants, simulated pre-tests included, begins before approval. Every participant gives written, explicit consent.

7. Review points

Two review points are proposed to the advisor: before the Statement of Work (13 November 2026) for the feasibility results, scenario coverage, the avatar method and its effort for the 300-sign library, the recording decision and the O4 fallback; and at the end of January 2027 for baseline and model accuracy. If a point is not met, the scope is narrowed with the advisor; no target is changed here.

Resource Requirements

Table 5 – Resources

Type	Requirement
Hardware	A GPU for training (the university GPU server if available, otherwise Google Colab, Kaggle or team members' GPUs; landmark models are light). For testing: a 1080p webcam, two Android phones, one iPhone and laptops.
Software	Python, PyTorch, MediaPipe, ONNX Runtime Web / TensorFlow.js, FastAPI, React/Next.js, Three.js, Blender, Docker, Jira, Confluence, GitHub and a design tool for prototypes (all free or with student licences).
AI services	API credits for the LLM, speech-to-text and text-to-speech services; otherwise open-source alternatives (local Whisper, an open-source LLM, Piper TTS).
Data	The Dataset (Methods 1): AUTSL (academic, non-commercial use; downloaded by a team member as reported, sample files verified, full dataset not yet checked) or BosphorusSign22k (licence agreement with its developers at Boğaziçi University); the Current TİD Dictionary as a reference (terms of use to be checked). No supplementary sign recordings are planned at present (conditions in Methods 1).
People	Desirable but not yet confirmed: a TİD interpreter as language consultant and Deaf, speech-impaired and blind volunteers via the university's disability support unit (initial contact made, response pending) and local associations; otherwise dictionary and Dataset references and simulated pre-tests only, reported as such.

Project Timeline

Table 6 – Work plan (October 2026 – June 2027)

Work package	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Requirements, literature review, ethics committee application	■	■							
Ethics committee review (expected 2–3 months)		■	■	■					
User research, interface prototypes		■	■						
Dataset access and audit, lexicon (MVP vocabulary + 300 signs)		■	■	■					
Feasibility check (data, coverage, landmarks, device benchmark)	■	■							
Recognition model v1 → v2			■	■	■				
Avatar rig and animation library			■	■	■	■			
Speech, LLM and TTS pipeline, back-end			■	■	■				
Front-end and accessibility layer				■	■	■			
MVP integration and internal testing						■	■		
WCAG audit, pre-tests, target-user evaluation							■	■	
Post-implementation report, demo video, poster, presentation								■	■

Use-Case Point Complexity

Following the Guidelines and the Statement of Work template, Table 7 declares the technical complexity factors we consider important for the project; the final decision rests with the reviewers.

Table 7 – Technical complexity factors

Factor	Description	Considered	Factor	Description	Considered
T1	Distributed system	No	T8	Portability to other platforms	Yes
T2	Response time / performance objectives	Yes	T9	System maintenance (deployment)	No
T3	End-user efficiency	Yes	T10	Concurrent / parallel processing	Yes
T4	Internal processing complexity	Yes	T11	Security features	Yes
T5	Code reusability	No	T12	Access for third parties (APIs)	Yes
T6	Easy to install	Yes	T13	End-user training	No
T7	Easy to use	Yes			

Importance and justification. *High:* T2 – median latency ≤2.0 s / ≤2.5 s and ≥15 fps landmark processing on a mid-range laptop (O1, O5); T3 – replaces interpreter booking and handwritten notes, with one-tap modes and quick phrases; T4 – landmark-based sign classification, LLM and rule-based gloss conversion, avatar clip blending; T7 – WCAG 2.2 AA, screen-reader, keyboard and voice control, usable without training (O6, O7); T8 – one PWA on Android, iOS and desktop browsers; T11 – on-device video processing, no raw video stored or sent, KVKK-compliant consent (O8). *Average:* T6 – a PWA opened from a link with no app store or setup; T10 – camera, recognition and speech pipelines run concurrently with asynchronous WebSocket messaging; whether landmarks or audio are streamed depends on Methods 6; T12 – external speech, LLM and TTS services behind one interface with open-source fallbacks. T1 is declared not important: a client–server PWA with one back-end and no distributed system of our own; reviewers may decide otherwise.

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